Dynamic Dark Energy from an Evolving Availability Reservoir

Solvency Field Theory (SFT) unified arXiv note — hazard exit + DESI posterior matching + two-channel closure + internal bandwidth motivation

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This note is a complete, self-contained presentation of the Solvency Field Theory (SFT) pipeline and its latest closure breakthrough. It supersedes prior incremental versions by (i) formalizing the hazard-exit ledger, (ii) documenting the DESI DR2/Y3 chain scoring method, (iii) replacing the constant closure α with an evolving two-channel availability reservoir U(z), and (iv) providing ablation and invariance checks that isolate the essential mechanism driving dynamical dark energy behavior.

Abstract

We propose a minimal emergent framework (Solvency Field Theory, SFT) that links cosmic acceleration to thermodynamic bookkeeping under a solvency constraint. A stable stock-and-hazard exit law replaces earlier shift-copy dropouts: $dS_{stock}/dt = \dot{S}_{in} - \Xi H(z) S_{stock}$, where Ξ is an effective exit strength. We combine IllustrisTNG (TNG300) telemetry as an income proxy with DESI DR2/Y3 Cobaya chains to score model predictions in the CPL plane (w0, wa) via a weighted KDE. Prior versions matched the DESI posteriors using a constant normalization α; here we eliminate this constant by introducing an evolving availability reservoir U(z) with a Goldilocks hump (fuel dominated) and a late-time geometric sustain channel (geometry dominated), smoothly gated by Γ(z). For time-varying U, internal consistency requires $I_{total}=\int U \cdot s_{stock}\, dt$ and a numerator $\dot{s}_{total}=U \cdot \dot{s}_{stock}+\dot{U} \cdot s_{stock}$. We theoretically ground this geometric channel in the thermodynamic processing limits of spacetime (the Margolus-Levitin bound) and local gravitational bandwidth saturation. The empirically required $\dot{U}$ term physically represents the 'Cosmic Echo'—the necessary expansion of the geometric spatial ledger to prevent a universal thermodynamic time-freeze as baryonic fuel engines fail. With this closure, best-fit points land in the DESI-favored quadrant (w0>−1, wa<0) for DESY5SN(zmin0.1), Pantheon+, and Union3 and achieve high KDE scores. Ablation tests show removing the $\dot{U}$ term collapses the posterior score by tens of log-units (and hundreds for inconsistent numerators), and a scale-invariance test confirms that the absolute amplitude of U(z) is a gauge under the consistent formulation.

1. Rules of construction

We enforce: (i) no new particles; (ii) prefer emergent/inevitable mechanisms; (iii) clearly separate derived relations from closures/calibrations.

2. SFT spine: mapping ledger dynamics to w(t) (derived)

We use the SFT equation-of-state mapping:

```
I(t) = ∫^t S_total(t′) dt′
w(t) = −1 − Ṡ_total(t) / [4π G ρ(t) I(t)]
```

All remaining work specifies S_total and Ṡ_total from a minimal ledger model and identifies ρ(t) (we report results for ρ=ρ_crit and ρ=ρ_m as sensitivity checks).

3. Exit term: from shift-copy failure to hazard exit (validated)

A shift-copy dropout Q̇_net(z)=Q̇_in(z)−Q̇_in(z_emit) collapses for monotone income histories (for any γ>1, the shifted term samples earlier, higher income). We therefore adopt a stock-and-hazard ledger:

```
dS_stock/dt = Ṡ_in(t) − λ(t) S_stock(t)
λ(z) = Ξ · H(z)
Ṡ_net(t) = Ṡ_in(t) − Ξ H(z) S_stock(t)
```

Crossing-only constraints produce a ridge in (γ,κ) because λ(z)=κH(z)ln(γ); the comparatively constrained combination is Ξ=κ ln(γ). We treat crossing ridges as a diagnostic and use full posterior scoring for parameter identification.

4. Inputs: TNG telemetry and DESI chains

Telemetry: IllustrisTNG TNG300 group catalogs at snapshots 33, 40, 50, 59, 63, 67, 72, 78, 84, 99 (z≈0–2).

```
L_SF ≈ (10^10 L_☉) · SFR
P_AGN ≈ ε (ṁ_BH c^2),  ε≈0.1
Ṡ_in ≈ (L_SF + P_AGN)/T_sink  (effective sink temperature T_sink used as a
fixed scale)
```

Observation: DESI DR2/Y3 Cobaya chain products under CPL w(z)=w0+wa·z/(1+z) for three SN combinations: DESY5SN(zmin0.1), Pantheon+, Union3.

5. DESI posterior scoring (KDE) — reproducible likelihood proxy

For each dataset, we construct a weighted 2D Gaussian KDE over chain samples (w0, wa). For each model realization we compute w(z), fit CPL (w0, wa) over z≤2, and score by KDE log-density (logKDE). Higher logKDE indicates closer agreement with the DESI posterior mass.

6. The closure breakthrough: two-channel availability reservoir U(z)

We replace the constant closure α with an evolving availability reservoir U(z) motivated by an 'S-curve' interaction picture: early constrained regime, a Goldilocks interaction window, then a bent late-time leg as geometric/horizon effects dominate.

```
Fuel hump:        U_fuel(z) = U0 (1+z)^m exp[−q(1+z)],   q = m/(1+z_peak)
Geometric sustain U_geom(z) = U_g
Gate:            Γ(z) = 1 / (1 + exp((z − z_g)/Δ))
Total:           U(z) = (1−Γ) U_fuel(z) + Γ U_geom(z)
```

Consistency for time-varying U:

```
s_stock = S_stock/V_phys ,   ṡ_stock = Ṡ_net/V_phys
s_total = U(z) · s_stock
I_total = ∫ s_total dt = ∫ U(z(t)) s_stock(t) dt
ṡ_total = d(s_total)/dt = U·ṡ_stock + U̇·s_stock    (mandatory)
```

The U̇ derivative is not merely a free parameter; it is the mathematical enforcement of Solvency Field Theory's core postulate. As local black hole event horizons fully saturate the spatial ledger's thermodynamic bandwidth, and global baryonic engines crash, the universe must shift the thermodynamic load. The `U̇·s_stock` term isolates this exact mechanism: the macroscopic expansion of the universal spatial ledger required to handle the thermodynamic debt.

7. Results (ρ=crit, FULL model)

Refined best-fit points from local zoom search (CPL fit over z≤2):

Dataset	logKDE	w0	wa	Ξ	m	z_peak	z_g	Δ	R=Ug/U0
DESY5SN(zmin0.1)	2.7519	-0.7906	-0.7606	5.500	3.0	1.368	0.609	0.271	1.180
Pantheon+	3.4274	-0.8455	-0.5963	5.674	4.0	1.162	0.397	0.162	0.665
Union3	2.8437	-0.6634	-1.0796	13.150	5.0	1.076	0.840	0.033	0.235

8. Internal correctness: ablation and invariance tests

8.1 Ablation: removing U collapses the fit

Dataset	Mode	logKDE	ΔlogKDE vs FULL	w0	wa
DESY5SN(zmin0.1)	FULL	2.754	0.000	-0.7915	-0.7550
DESY5SN(zmin0.1)	NO_UDOT	-33.970	-36.725	-0.8009	-0.2328
DESY5SN(zmin0.1)	HACK	-391.760	-394.514	-0.6608	-0.3414
Pantheon+	FULL	3.427	0.000	-0.8448	-0.5999
Pantheon+	NO_UDOT	-24.393	-27.820	-0.8373	-0.1826
Pantheon+	HACK	-241.743	-245.170	-0.7739	-0.2151
Union3	FULL	2.845	0.000	-0.6634	-1.0774
Union3	NO_UDOT	-2.173	-5.017	-0.8228	-0.3530
Union3	HACK	-1260.341	-1263.186	-0.1417	-1.7048

8.2 Scale invariance: U amplitude is a gauge under consistent numerators

Dataset	Mode	Range of logKDE across U-scale factors
DESY5SN(zmin0.1)	FULL	0.00e+00
DESY5SN(zmin0.1)	NO_UDOT	0.00e+00
DESY5SN(zmin0.1)	HACK	1.02e+05
Pantheon+	FULL	0.00e+00
Pantheon+	NO_UDOT	3.91e-14
Pantheon+	HACK	1.03e+05
Union3	FULL	0.00e+00
Union3	NO_UDOT	0.00e+00
Union3	HACK	3.79e+05

9. Robustness bands from local zoom samples

Dataset	Band	Peak	N	Ξ	z_peak	z_g	Δ	R	w0 range	wa range
DESY5SN(zmin0.1)	Δ≤0.1	2.752	220	4.50–8.50	0.88–1.84	0.00–0.87	0.040–0.765	0.342–3.083	-0.834–-0.749	-0.861–-0.592
DESY5SN(zmin0.1)	Δ≤0.5	2.752	1168	4.50–8.50	0.84–2.08	0.00–1.08	0.027–1.000	0.289–3.717	-0.899–-0.685	-1.059–-0.401
Pantheon+	Δ≤0.1	3.427	150	1.71–5.67	0.89–2.03	0.00–1.02	0.063–0.672	0.436–2.762	-0.862–-0.811	-0.699–-0.519

Pantheon+	Δ≤0.5	3.427	739	1.71–5.67	0.76–2.25	0.00–1.07	0.029–1.000	0.395–3.493	-0.897–-0.785	-0.806–-0.389
Union3	Δ≤0.1	2.844	359	10.75–14.75	0.39–1.59	0.50–1.73	0.009–0.227	0.035–1.156	-0.712–-0.632	-1.185–-0.945
Union3	Δ≤0.5	2.844	1611	10.75–14.75	0.20–1.59	0.21–2.13	0.009–0.392	0.024–1.156	-0.760–-0.583	-1.353–-0.768

10. Physical motivation (internal notes)

The closure U(z) is presented as an effective parameterization; the following motivates why an evolving reservoir and a U term are physically plausible under SFT's 'finite processing bandwidth' interpretation.

- Proper time as local exhaust: proper time τ can be interpreted as accumulated irreversible internal processing, bounded by quantum limits (e.g., Margolus–Levitin).
- GR as a bandwidth sink: maintaining curved geometry near mass concentrates bookkeeping overhead, throttling locally available internal evolution; this maps onto gravitational time dilation for stationary observers.
- Black holes as saturation: horizons represent limiting cases where geometry maintenance dominates; in observer-dependent terms this motivates a 'geometric takeover' channel.
- Cosmic-scale echo: the late-time geometric sustain channel and the empirically mandatory U·s_stock term are the global analogue of local horizon/bandwidth behavior (the 'bent S-curve back leg').

11. Relationship to earlier SFT versions

Earlier versions established the hazard exit law and demonstrated full DESI posterior matching with a constant closure α. The v6 unified note subsumes these as special cases: constant-U implies U=0 and recovers the constant-α closure.

12. Reproducibility and artifacts

To reproduce the results in this note, we recommend packaging the following artifacts as ancillary files:

- telemetry_supermaster.csv
- DESI Cobaya chain folders (chain.*.txt, input yaml, etc.) for the three datasets
- twochannel_v2_bestfits_summary.csv (coarse scan bestfits)
- v2p1_ablation_summary.csv
- v2p1_scale_invariance_check.csv
- v2p1_bestfit_refined_summary.csv

- v2p1_zoom_top_*_rho_crit.csv (optional but helpful for robustness bands)
- Figure pack PNGs (w0–wa scatter, ablation bars, scale-invariance plots)

13. Limitations and next steps

- **Phase II Micro-Derivation:** This paper establishes the macroscopic mathematical necessity of the geometric expansion channel (U(z)) and its physical motivation (Gravitational Bandwidth Saturation). The immediate next step for SFT is to dynamically derive this exact curve from the micro-physics. Future work will replace the parametric Γ(z) gate by directly integrating the physical Bekenstein-Hawking event horizon surface areas $\sum 16\pi G^2 M^2/c^4$ from the TNG telemetry, proving that microscopic black hole data natively generates the macroscopic expansion observed by DESI.

- Extend z_peak range and refine local scans to sharpen inferred hump location.
- Cross-validate on alternative simulation suites/volumes; propagate uncertainties into predicted w(z) bands.